



Unravel Carbon Empowers Organizations with a Climate Platform for Decision-Making

Learn how the Asia-headquartered climate technology company is using AWS technology to drive measurable outcomes for enterprises, while creating new business value.

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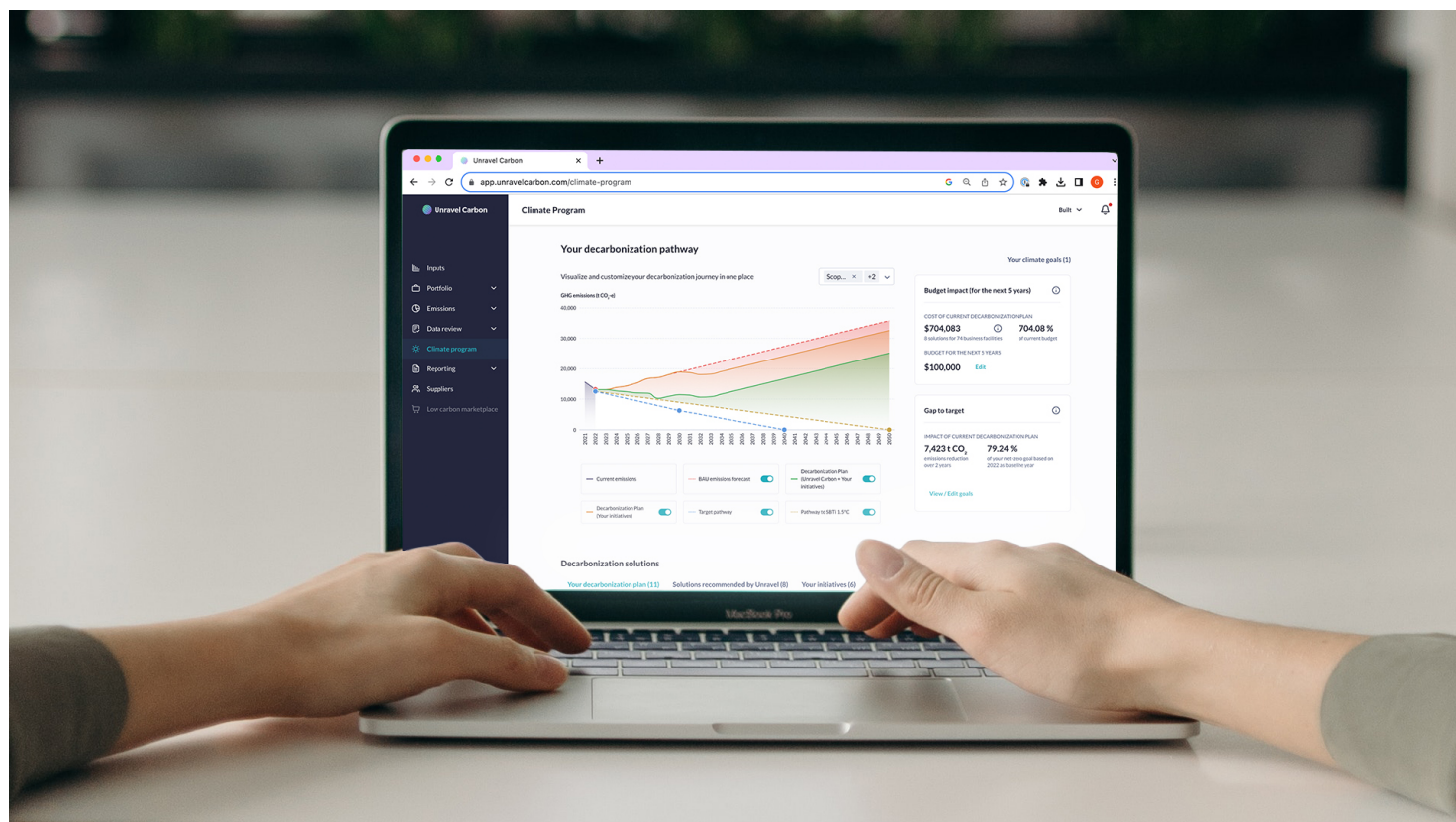
Overview

As the world grapples with the urgent need to address climate change, [Unravel Carbon](#) emerges as a beacon of action. Positioned as a decision-making climate platform, the company seeks to catalyze change with its team of climate experts, software engineers, and data scientists.

By providing credible and reliable data and insights for companies to track and reduce their environmental impact, Unravel Carbon has delivered remarkable results for its customers—from cost-effective scalability to helping them reduce emissions by up to 36 percent. The platform also automates much of the reporting process, saving an average of 50 to 60 percent of the total project time and manpower required.

Through its platform, built on AWS serverless architecture, the company converts accounting information into actionable carbon insights with the help of artificial intelligence (AI). This equips organizations to reduce business risks, proactively advance toward achieving net zero emissions, and explore new growth opportunities in the low-carbon economy.

In the short span of almost two years, the company has also expanded its operations into Australia & New Zealand, Southeast Asia, and Hong Kong.



Founded in 2022, Unravel Carbon embarked on a clear mission—to accelerate the world's progress toward a zero-carbon economy. Co-founders Grace Sai and Marc Allen, driven by concerns over escalating global temperatures and a commitment to secure a better world for future generations, looked to address Asia Pacific's carbon footprint. The region is home to 70 percent of the world's supply chains, and is responsible for 60 percent of global emissions.

Unravel Carbon aims to empower medium to large enterprises in effectively assessing, reducing, and reporting their carbon emissions, thereby enabling businesses to prepare and plan for risks and opportunities in the low-carbon economy. To assist organizations through their decarbonization journey, the company needed to create a user-friendly and scalable platform tailored to meet customer demands. As a startup, Unravel Carbon prioritized the agility to establish and expand quickly, the flexibility to experiment, and cost efficiencies.

Crucially, data security was imperative as the company would be dealing with sensitive financial information. Heading into 2024, more enterprises in Asia Pacific will face mandates to comply with and incorporate [Scope 3](#) reporting standards. For enterprises entrusting sensitive data to the Unravel Carbon platform, working with an ISO 27001 and SOC 2 certified company that prioritizes data security, management, and compliance, is paramount.

Transitioning to AWS proved transformative. In 2022, the company launched its enterprise-grade Software-as-a-Service solution hosted on AWS.

Solution | Driving Decarbonization through Data Intelligence

[AWS Lambda](#), a serverless, event-driven compute service, lies at the core of Unravel Carbon's platform. This architecture provides the company the flexibility and scalability to efficiently handle peak compute demands while optimizing costs. By averting the operation and upkeep of idle on-premises resources and the need for a full-time maintenance engineer, the company saves US\$100K in annual operating costs.

Additionally, Unravel Carbon deploys [AWS Fargate](#), a serverless container management service, to manage the orchestration tool in the company's data infrastructure. This tool helps schedule the company's data pipelines more effectively, significantly reducing the time and effort needed to manage its infrastructure by 50%. With increased operational efficiency, Unravel Carbon can more effectively serve its customers.



To safeguard sensitive enterprise data, including financial information, the company uses [Amazon Virtual Private Cloud](#) (Amazon VPC), a cloud computing service that creates logically isolated virtual networks. With this, Unravel Carbon ensures that access, connectivity, and compute resource placement are assigned according to specific customer profiles while maintaining stringent security protection and compliance.

Unravel Carbon has also received both training and resources through the [AWS Startup Ramp](#) to support its early-stage growth, including cost optimization workshops. The company has actively participated in various partner initiatives and programs within the AWS Partners Network. This has helped to drive Unravel Carbon's go-to-market strategy and rapid growth in the region.

Outcome | Advancing Climate Goals and Innovation

Unravel Carbon's collaboration with AWS has catalyzed a promising transformation in the decarbonization landscape, exemplifying both companies' commitment to sustainability and innovation. Amazon has itself made the Amazon Climate Pledge to reach net zero emissions by 2040, a decade ahead of the Paris Agreement.

Looking ahead, Unravel Carbon will focus on further development and integration of quantitative analytics and Machine Learning, further enhancing flexible data ingestion and processing, and streamlining reporting with generative AI (GenAI) assistance.

"The next frontier of the climate change challenge is one of data management. Few companies can move the needle because existing emissions measuring and reporting processes are slow, laborious, and lack actionable insights. Through a working product built on AWS, we demonstrate tangible, measurable, and impactful change—making decarbonization a feasible and compelling reality," said Grace Sai, CEO & Co-Founder, Unravel Carbon.

Learn More

To learn more, visit aws.amazon.com/sustainability.

About Unravel Carbon

An AWS Partner, Unravel Carbon is an enterprise software company that helps companies measure, track and reduce their carbon emissions, focusing on Asia Pacific's supply chains. It is a leading AI-powered decarbonization platform in Asia that converts any company's accounting data into full supply-chain carbon data in weeks, providing detailed emission analytics, and relevant climate solutions. The Singapore-headquartered company also operates across Australia & New Zealand, Southeast Asia, and Hong Kong.

AWS Services Used

AWS Lambda



AWS Lambda is a compute service that runs your code in response to events and automatically manages the compute resources, making it the fastest way to turn an idea into a modern, production, serverless applications.

[Learn more »](#)

AWS Fargate

AWS Fargate is a serverless, pay-as-you-go compute engine that lets you focus on building applications without managing servers.

[Learn more »](#)

Amazon Virtual Private Cloud

Amazon Virtual Private Cloud (Amazon VPC) gives you full control over your virtual networking environment, including resource placement, connectivity, and security.

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Amazon S3

Amazon Simple Storage Service (Amazon S3) is an object storage service offering industry-leading scalability, data availability, security, and performance.

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